

Emily R. Stanton

Principal AI Engineer & LLM Systems Architect

Contact Information

Email: emily.stanton.ai@example.com

Phone: +1 (415) 555-9023

Location: Seattle, WA, USA

LinkedIn: linkedin.com/in/emilystanton-LLM

GitHub: github.com/emily-ml

Professional Summary

Principal AI Engineer with 10+ years of experience designing large-scale AI and RAG systems at major technology companies including Google, Meta, and Microsoft. Expert in LLM architecture, retrieval systems, distributed training pipelines, and high-performance inference. Proven success leading engineering teams to build reliable AI solutions serving millions of global users.

Education

Ph.D. in Machine Learning – MIT (2014–2018)

Bachelor of Science in Computer Science – Carnegie Mellon University (2010–2014)

Technical Skills

- Programming: Python, C++, Go, Rust (intermediate), TypeScript
- Deep Learning: PyTorch, TensorFlow, JAX, Flax, Haiku
- LLM & RAG: GPT models, PaLM, Llama 3, Mistral, Hybrid Search, Distributed FAISS, Pinecone
- Cloud: GCP (expert), AWS, Azure, Kubernetes, Borg
- MLOps: TFX, Vertex AI, MLflow, Ray Serve, KServe

- Data: BigQuery, Dataflow, Spark, Snowflake

Professional Experience

Principal AI Engineer – Google DeepMind (2022–Present)

- Led architecture of Google's internal RAG platform used across multiple product teams.
- Improved ranking precision by 27% using advanced embedding distillation techniques.
- Deployed LLM inference pipelines using JAX and TPU v5p with 30% latency reduction.
- Supervised 12 engineers working on hybrid retrieval and model evaluation systems.

Senior Machine Learning Engineer – Meta AI (2019–2022)

- Developed large-scale transformer-based relevance models for social graph ranking.
- Designed distributed FAISS indexing system handling billions of embeddings.
- Built LoRA-based fine-tuning workflows for Meta's internal LLMs.

Machine Learning Engineer – Microsoft Azure AI (2016–2019)

- Implemented neural ranking models for Azure Cognitive Search with measurable improvements.
- Contributed to Azure ML distributed training APIs and ONNX Runtime acceleration.

AI Research Intern – Amazon Alexa AI (2015)

- Improved speech-to-text transformer encoder resulting in 5% accuracy gain.

Projects

- Enterprise RAG system for large-scale search
- Multimodal LLM combining text, image, and code embeddings

- Distributed embedding factory for terabyte-scale datasets

Certifications

- Google Cloud Professional ML Engineer
- NVIDIA Deep Learning Institute Optimization Cert.

Publications

- “Hybrid Retrieval at Enterprise Scale,” NeurIPS Industry Track 2023
- “Sharding Transformers for Low-Latency Inference,” AAAI 2022

Awards

- Google DeepMind Outstanding Engineering Award (2023)
- Meta AI Impact Award (2021)

Languages

English (Native), French (Intermediate)